



Shenzhen Sipeeed Technology Co.,Ltd.

Sheet: /

File: Tang_Mega_138K_Console_32001C.kicad_sch

Title: Tang 138K Console V32001C

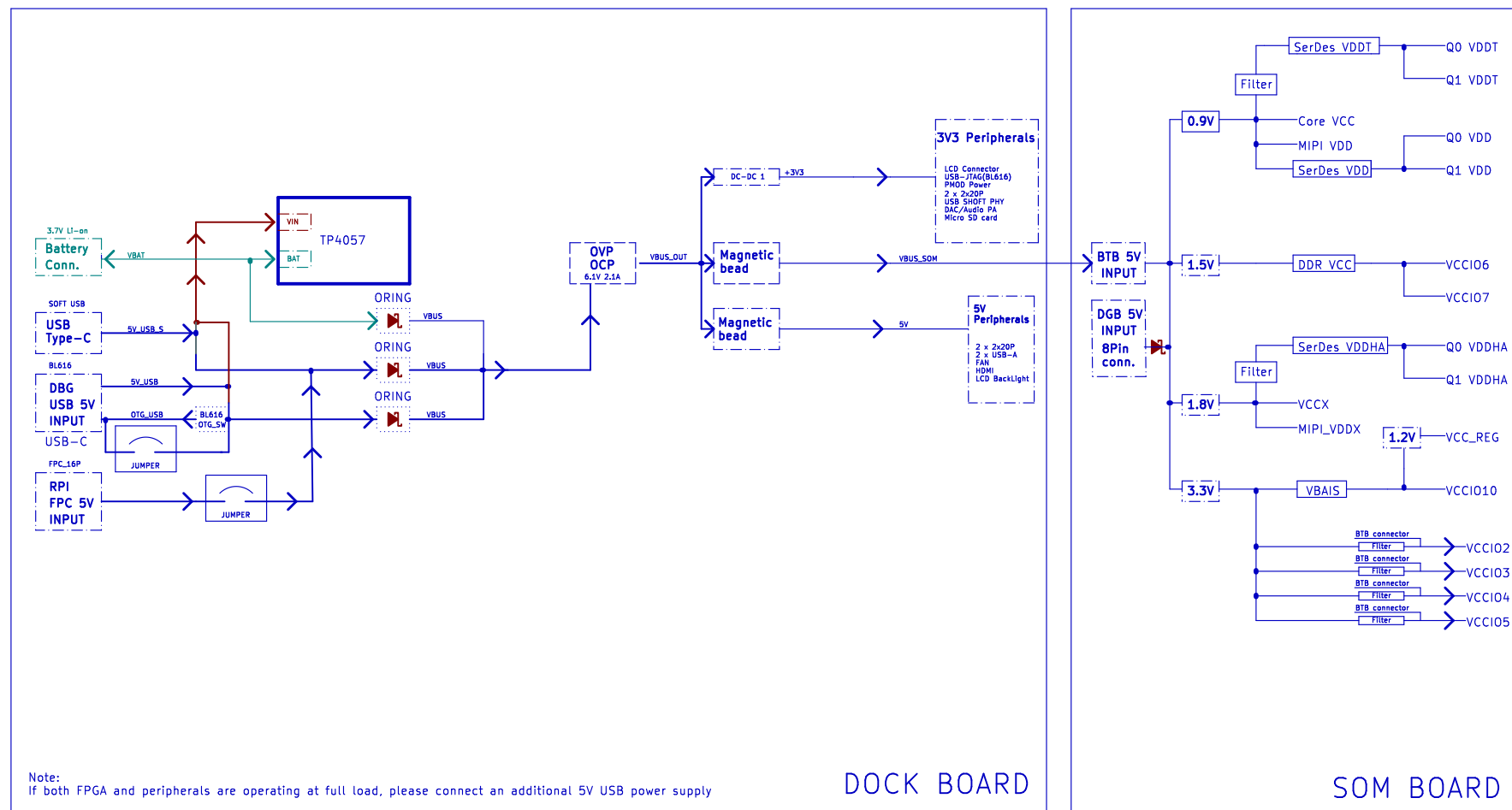
Size: A4	Date: 2025-04-24
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Size: A4 Date: 2025-04-24
KIC-LEDA-001

KIC80 E.D.A. 9.0.1

Rev: 1.3

Id: 1/17



A Retro Ready Console Powered by FPGA

Shenzhen Sipeed Technology Co.,Ltd.

Sheet: /POWER_TREE/

File: PWR_TREE.kicad_sch

Title: Tang_138K_Console_V32001C

Size: A4

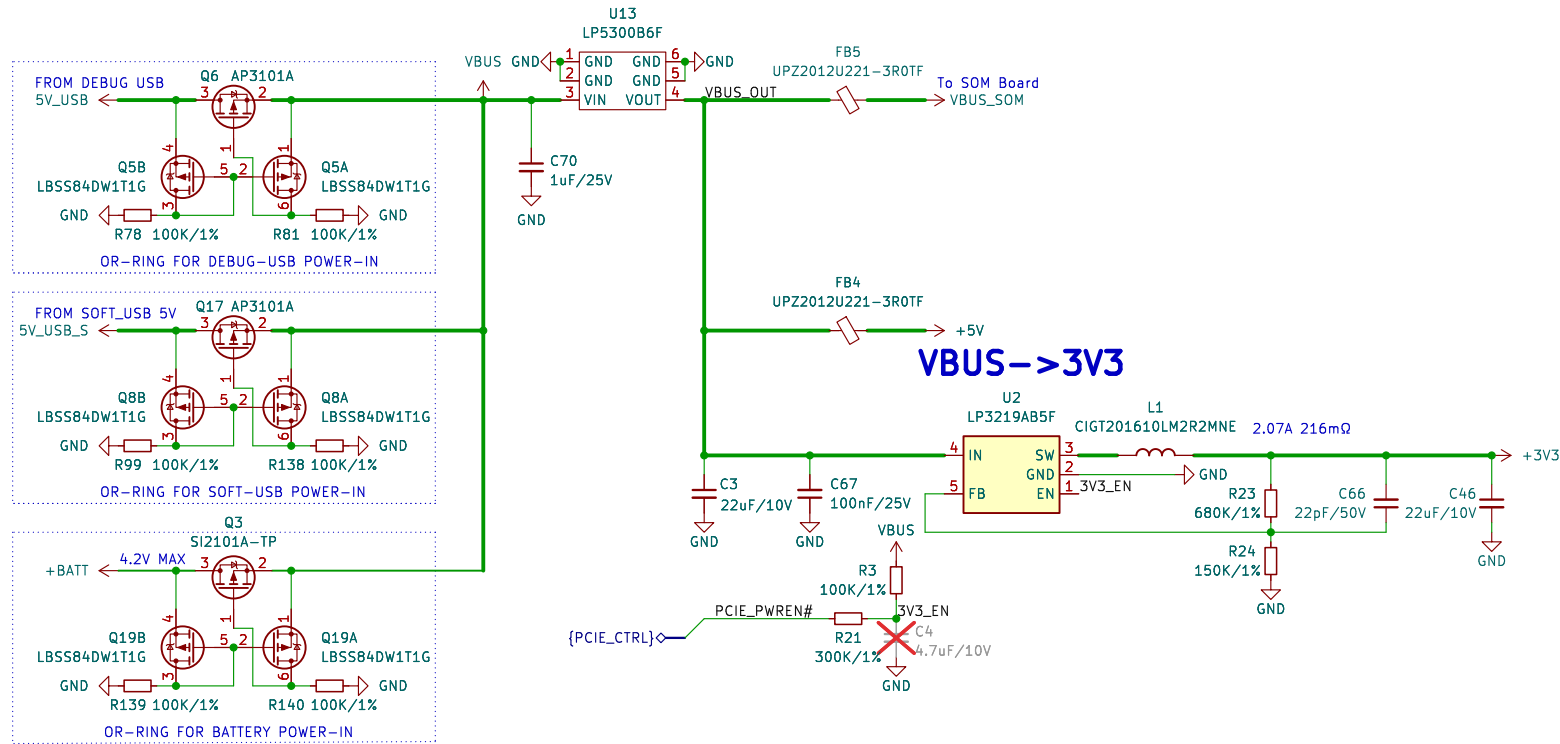
Date: 2025-04-24

Rev: 1.3

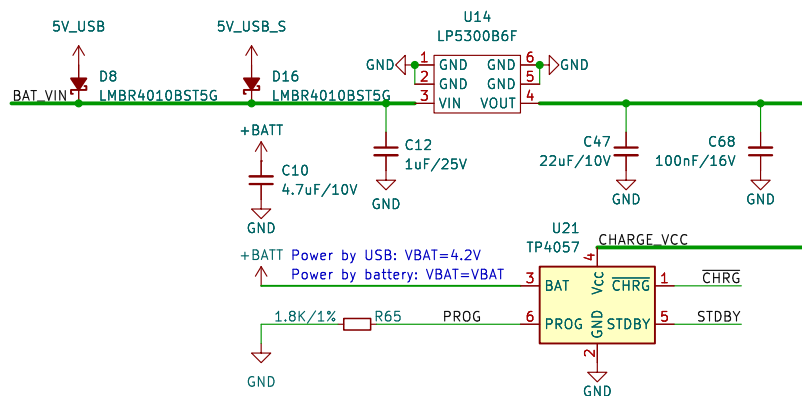
KiCad E.D.A. 9.0.1

Id: 2/17

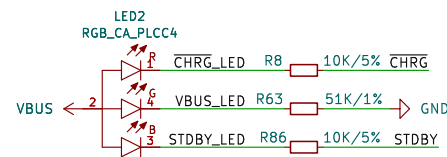
SYS POWER



Battery CHRG



VBUS_LED



Connect to
3.7V Li-on battery



A Retro Ready Console Powered by FPGA

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Sheet: /POWER/

File: PWR_MAIN.kicad_sch

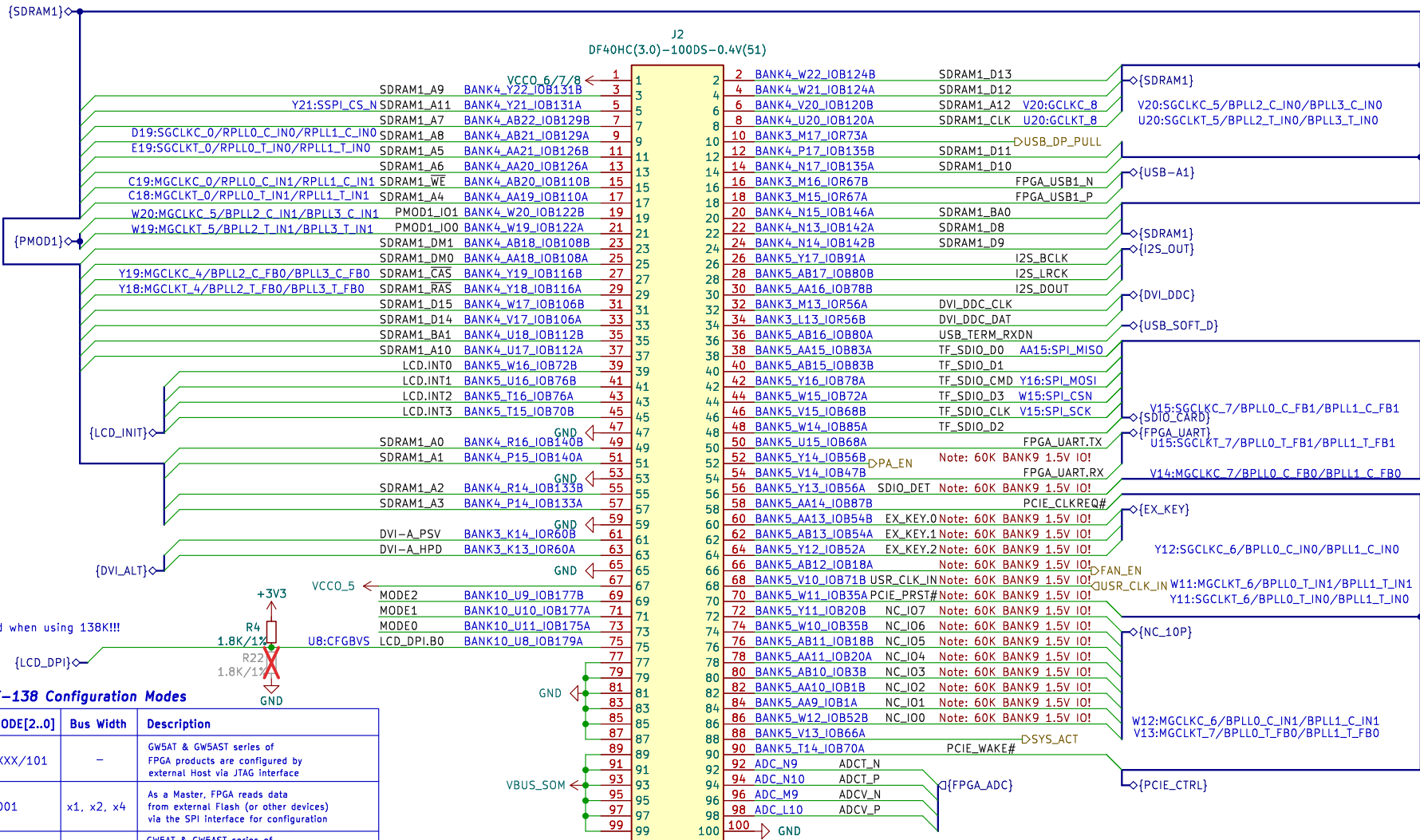
Title: Tang_138K_Console_V32001C

Size: A4 Date: 2025-04-24

KiCad E.D.A. 9.0.1

Rev: 1.3

Id: 3/17

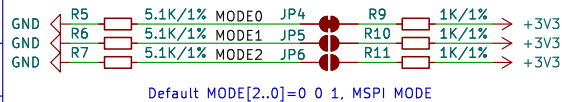


GW5AT-138/GW5AST-138 Configuration Modes

Configuration MODE	MODE[2..0]	Bus Width	Description
JTAG	XXX/101	—	GW5AT & GW5AST series of FPGA products are configured by external Host via JTAG interface
MSPI	001	x1, x2, x4	As a Master, FPGA reads data from external Flash (or other devices) via the SPI interface for configuration
SSPI	010	x1, x4	GW5AT & GW5AST series of FPGA products are configured by external Host via SPI interface
Master SERIAL	000	x1	Before FPGA used as a Slave FPGA reads data from external devices via the DIN interface for configuration
Slave SERIAL	111	x1	GW5AT & GW5AST series of FPGA products are configured by external Host via DIN interface
Master CPU	100	x8, x16, x32	Before FPGA used as a Slave FPGA reads data from external devices via the DBUS interface for configuration
Slave CPU	110	x8, x16, x32	GW5AT & GW5AST series of FPGA products are configured by external Host via DBUS interface

Please refer to GOWIN Manual UG984/UG984E for more details.

MODE Switch



A Retro Ready Console Powered by FPGA

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Sheet: /SOM_BT2/

File: SOM_BT2.kicad_sch

Title: Tang_138K_Console_V32001C

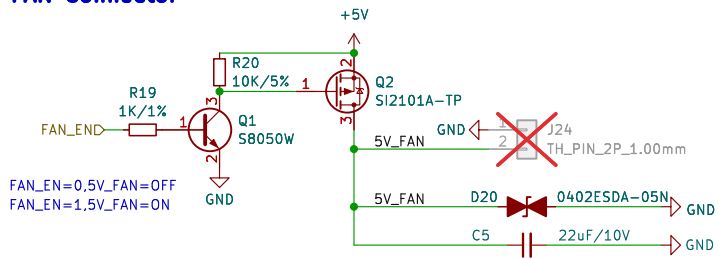
Size: A4 Date: 2025-04-24

KiCad E.D.A. 9.0.1

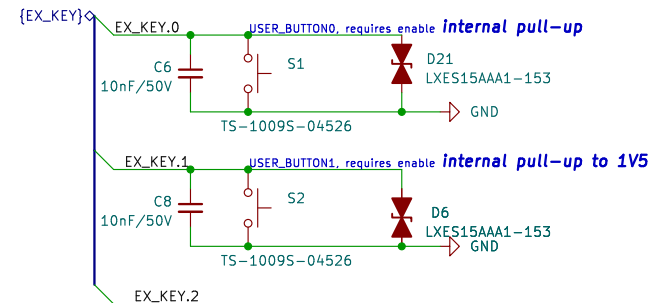
Rev: 1.3

Id: 6/17

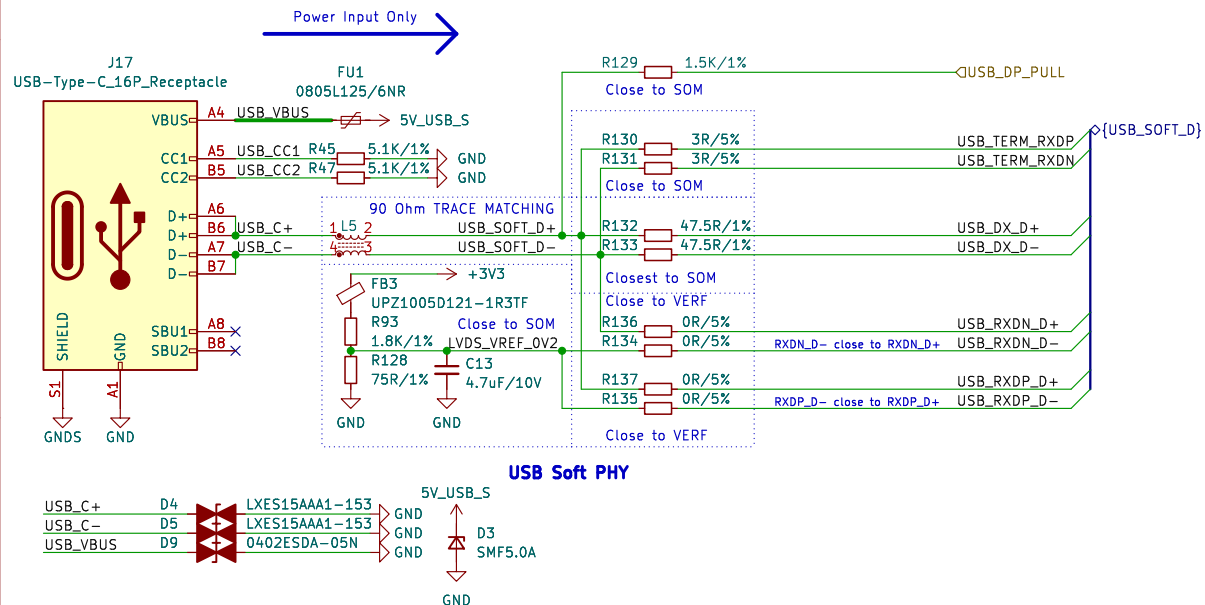
FAN Connector



BUTTON



USB Soft Deivce



A Retro Ready Console Powered by FPGA

Shenzhen Sipeed Technology Co., Ltd.

Sheet: /PERIPHERALS/

File: PERIPHERALS.kicad_sch

Title: Tang_138K_Console_V32001C

Size: A4

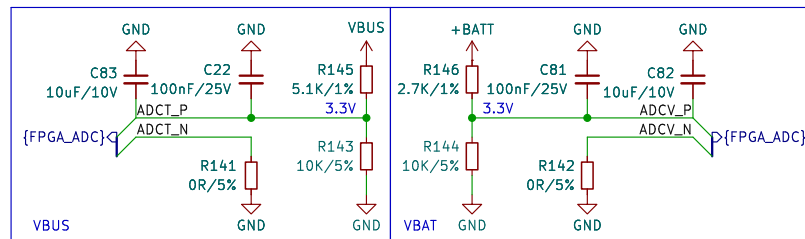
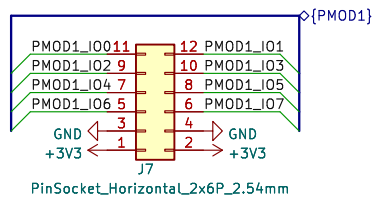
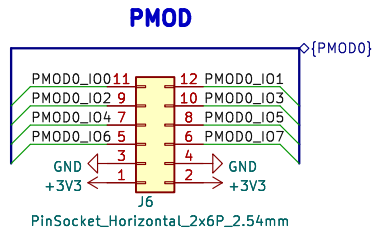
Date: 2025-04-24

KiCad E.D.A. 9.0.1

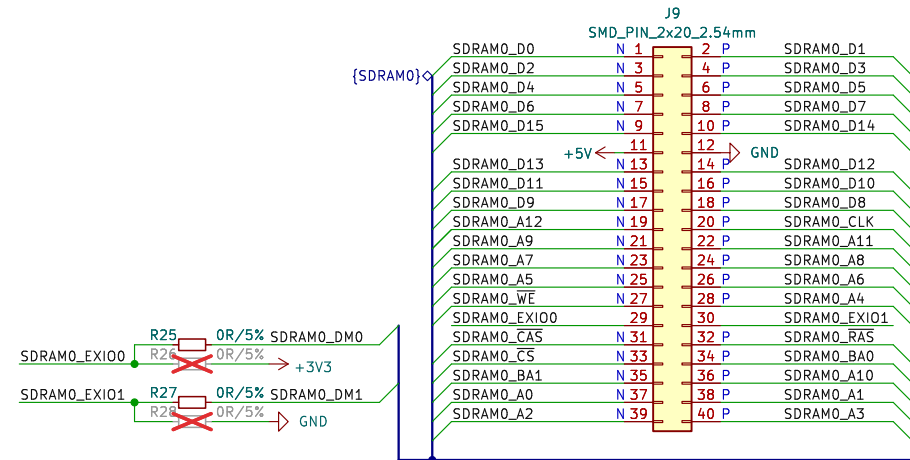
Rev: 1.3

Id: 7/17

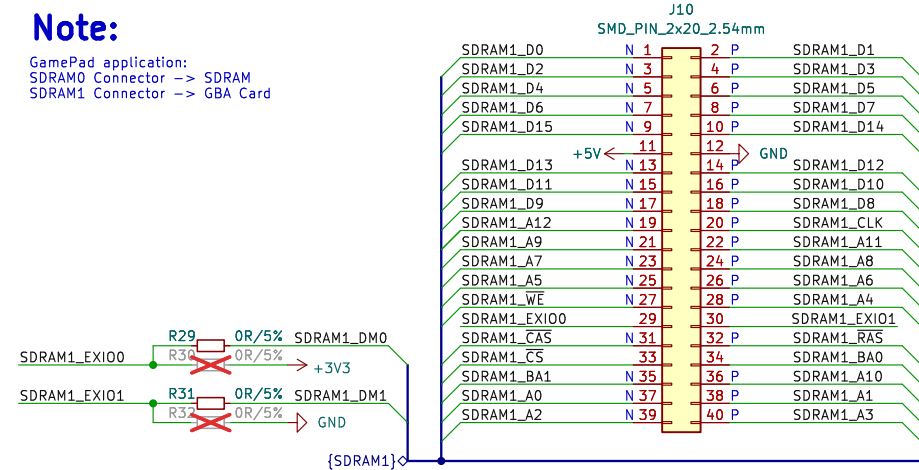
EX CONN.



SDRAM0 CONN. Close to PCIE connector



SDRAM1 CONN. Close to PMOD connector



Note:

GamePad application:
SDRAM0 Connector -> SDRAM
SDRAM1 Connector -> GBA Card



A Retro Ready Console Powered by FPGA

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Sheet: /FPGA_EXT_CONN/

File: EX_CONN..kicad_sch

Title: Tang_138K_Console_V32001C

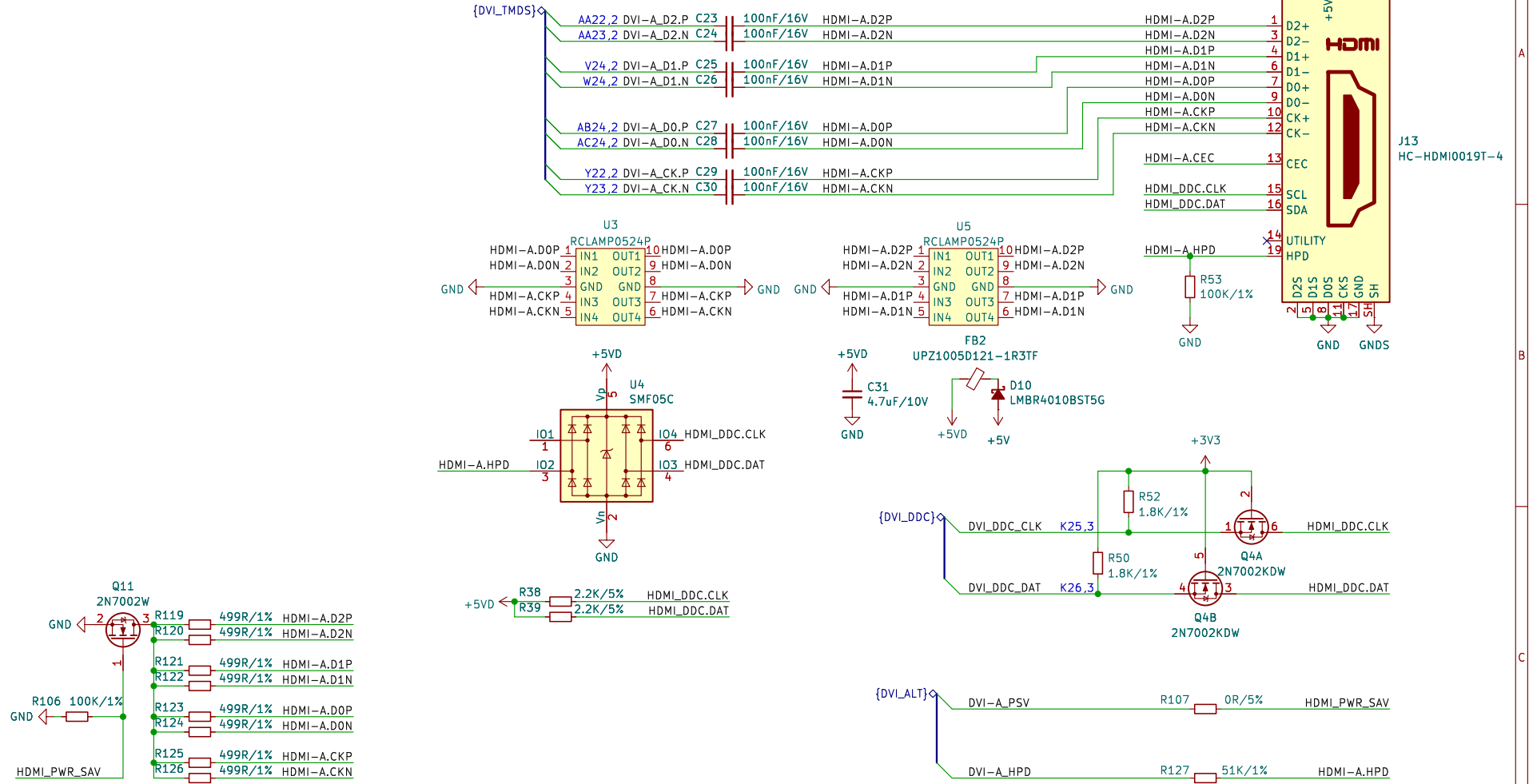
Size: A4	Date: 2025-04-24
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Rev: 1.3

Id: 8/17

DVI OUTPUT



A Retro Ready Console Powered by FPGA

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Sheet: /DVI_TMDS/

File: TMDS_TX.kicad_sch

Title: Tang_138K_Console_V32001C

Size: A4

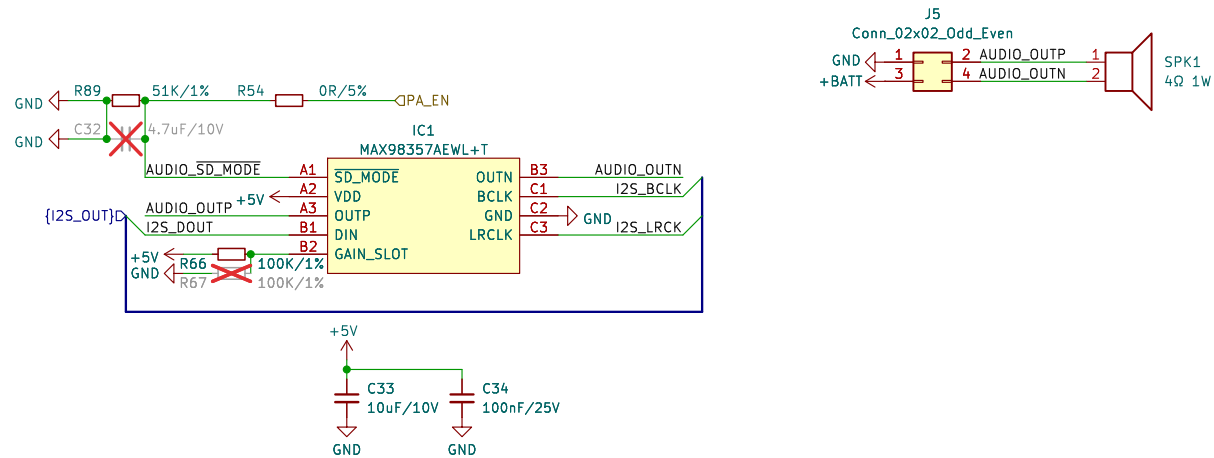
Date: 2025-04-24

Rev: 1.3

KiCad E.D.A. 9.0.1

Id: 9/17

DAC + Audio PA



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Sheet: /AUDIO+PA/
File: AUDIO.kicad_sch

Title: Tang_138K_Console_V32001C

Size: A4

Date: 2025-04-24

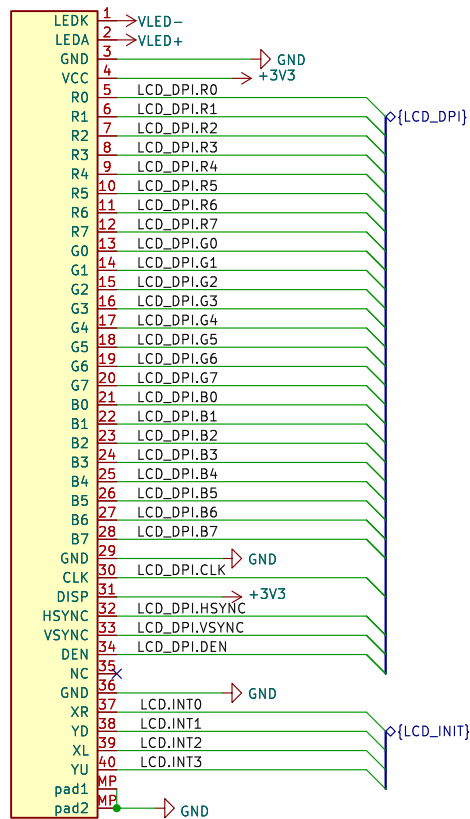
Rev: 1.3

KiCad E.D.A. 9.0.1

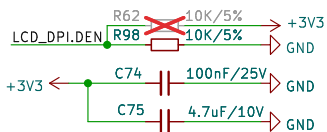
Id: 10/17

RGB LCD Display

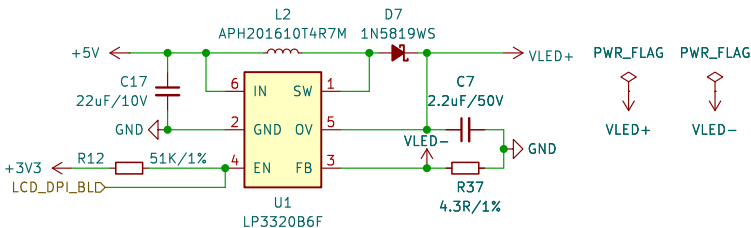
RGB LCD Display Connector



J3
FPC-40P-0.5MM-H2.0MM-rearlock



BACKLIGHT Driver



Feedback Voltage = 250mV
BackLight Current = $250\text{mV}/4.3\text{R} \approx 58.1\text{mA}$
7inch LCD QTM070TFT40-001 : IF=20*3=60mA

LCD.INT1 R13 5.1K/1% +3V3
LCD.INT0 R36 5.1K/1% +3V3



A Retro Ready Console Powered by FPGA

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Sheet: /LCD_DPI_24BIT/

File: LCD_DPI.kicad_sch

Title: Tang_138K_Console_V32001C

Size: A4

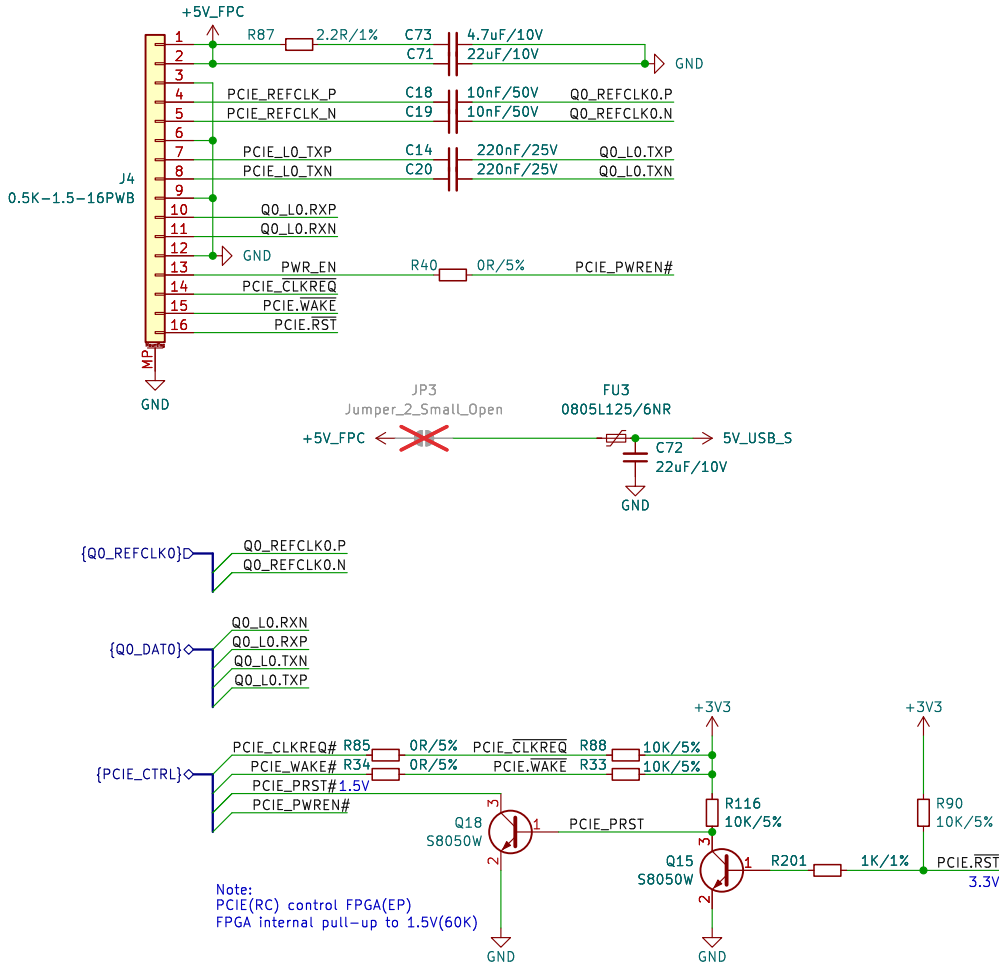
Date: 2025-04-24

Rev: 1.3

KiCad E.D.A. 9.0.1

Id: 11/17

RPI HAT+ PCIE



{Q0_REFCLK0}D

{Q0_DATA0}D

{PCIE_CTRL}D

Note:
PCIE(RC) control FPGA(EP)
FPGA internal pull-up to 1.5V(60K)



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Sheet: /PCIE_GT/
File: SERDES.kicad_sch

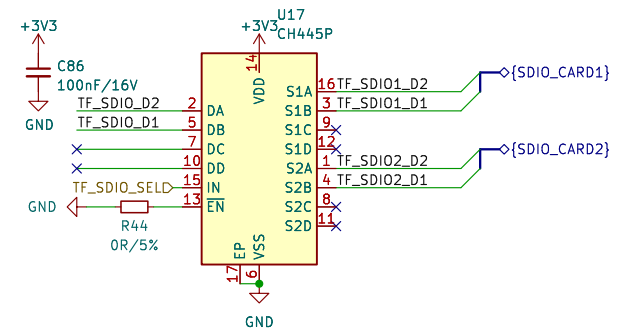
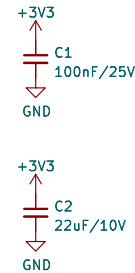
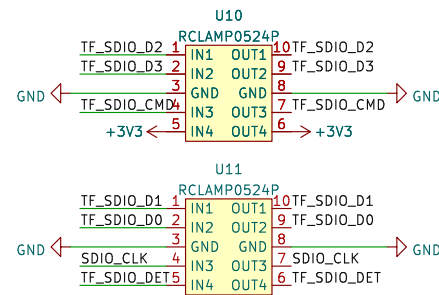
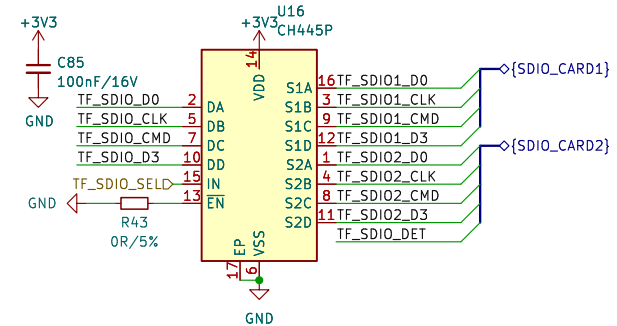
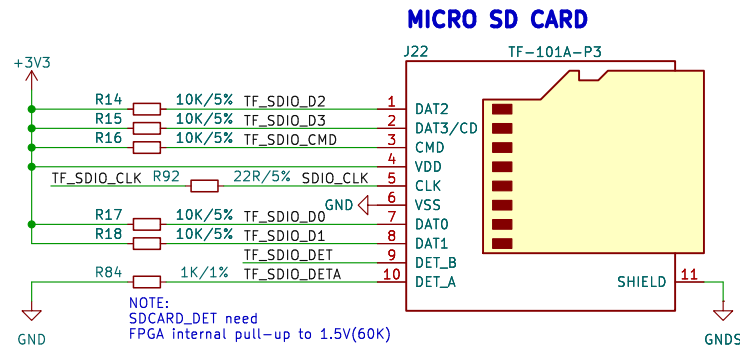
Title: Tang_138K_Console_V32001C

Size: A4 Date: 2025-04-24

KiCad E.D.A. 9.0.1

Rev: 1.3

Id: 12/17



A Retro Ready Console Powered by FPGA

Shenzhen Sippeed Technology Co., Ltd.

Sheet: /SDIO_CARD/

File: SDCARD.kicad_sch

Title: Tang_138K_Console_V32001C

Size: A4 Date: 2025-04-24

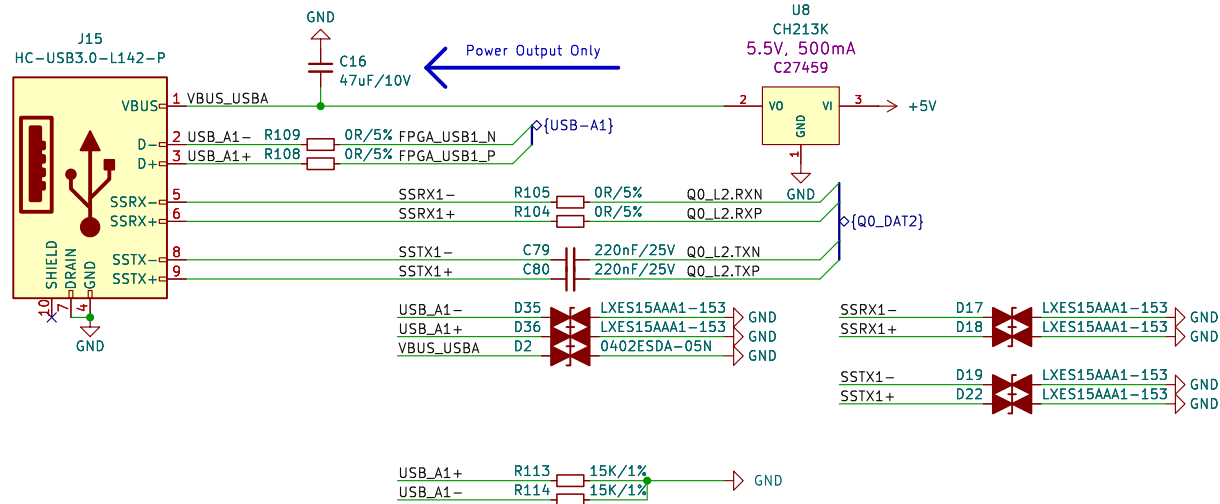
KiCad E.D.A. 9.0.1

Rev: 1.3

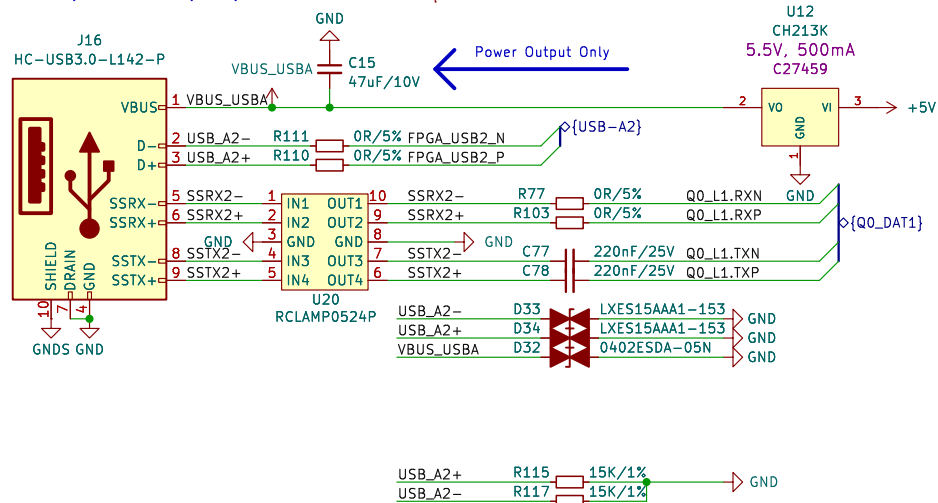
Id: 14/17

USB_Ax2

USB_A1 Full-Speed Host + Super-Speed Device



USB_A2 Full-Speed Host + Super-Speed Device



A Retro Ready Console Powered by FPGA

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Sheet: /USB3-A_GT/

File: USB3-GT.kicad_sch

Title: Tang_138K_Console_V32001C

Size: A4 Date: 2025-04-24

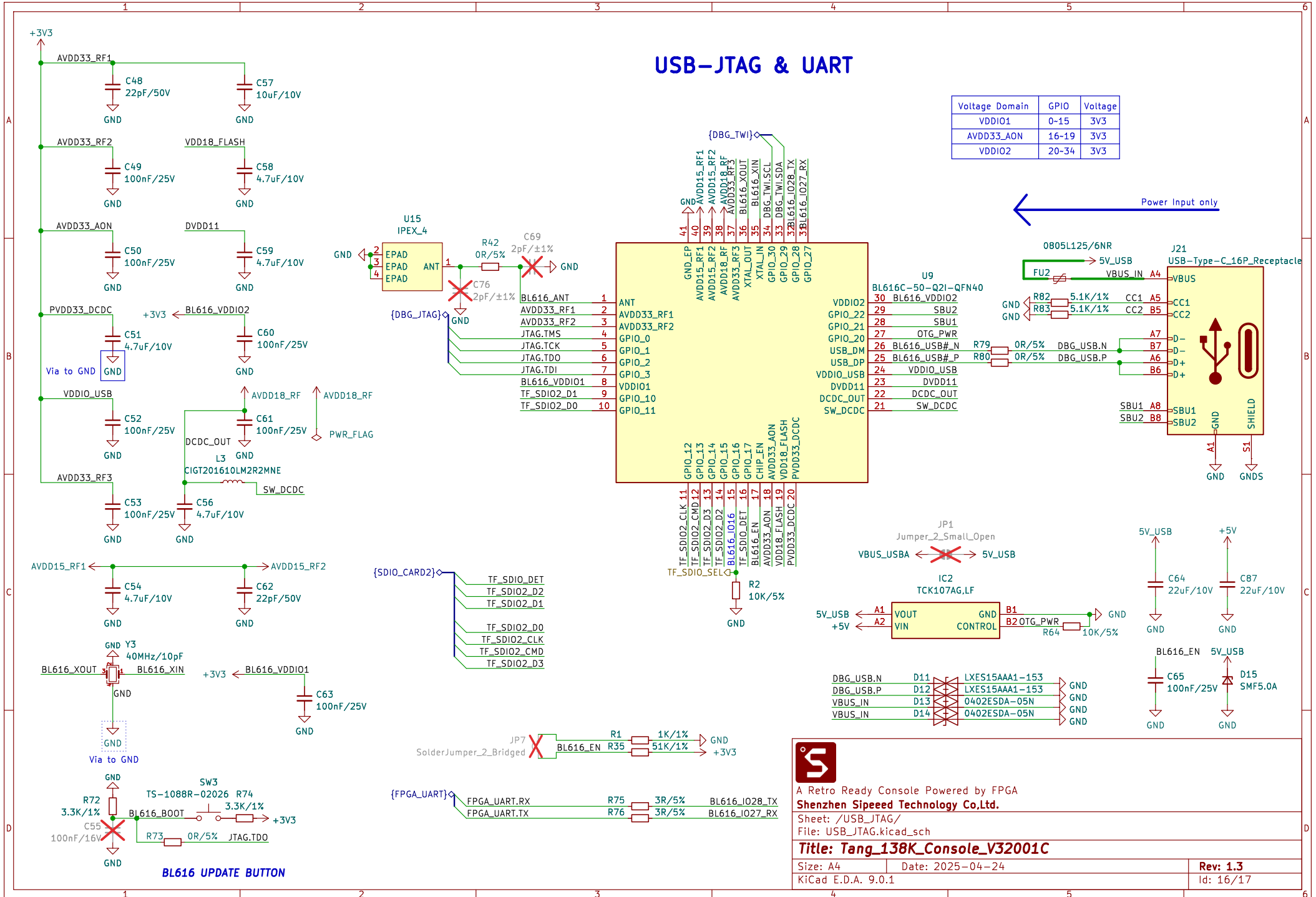
KiCad E.D.A. 9.0.1

Rev: 1.3

Id: 14/17

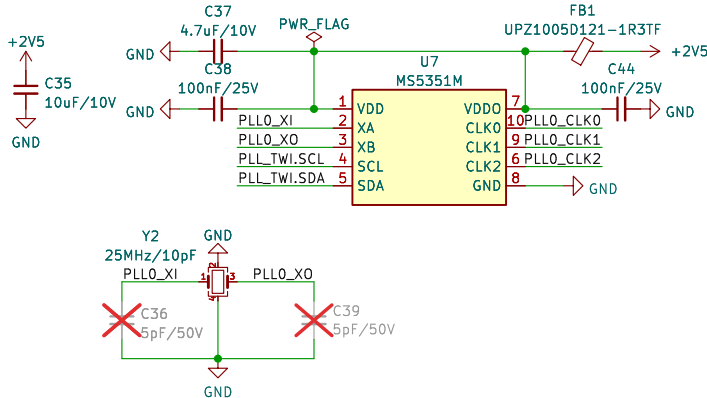
USB-JTAG & UART

Voltage Domain	GPIO	Voltage
VDDIO1	0-15	3V3
AVDD33_AON	16-19	3V3
VDDIO2	20-34	3V3

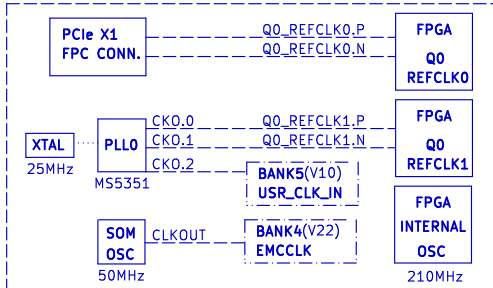


CLOCK

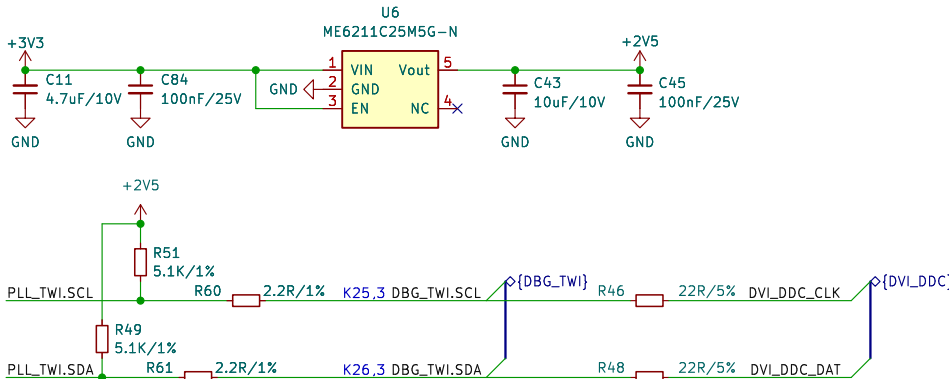
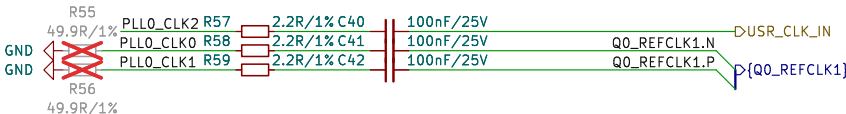
EXT. PLI



COLCK TREE



FPGA internal clock generator OSC.
The user can change the value of
the option "Frequency Divider" to
generate a different output clock
frequency, which can be one of
64 values: 1/2, 1/3, 1/4, 1/6, 1/8, 1/10,
..... or 1/126 of the oscillator frequency.



A Retro Ready Console Powered by FPGA

Shenzhen Sipeeed Technology Co.,Ltd.

Sheet: /SYS_CLOCK/

File: CLOCK.kicad_sch

Title: Tang_138K_Console_V32001C

Size: A4

Date: 2025-04-24

KiCad E.D.A. 9.0.1

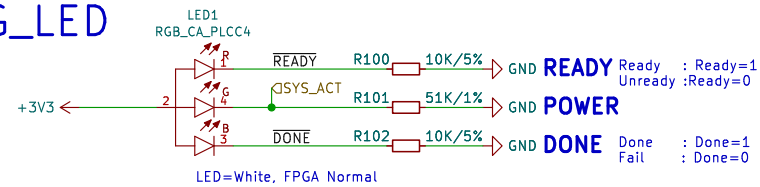
Rev: 1.3

Id: 18/17

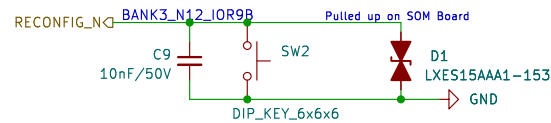
Inverter Circuit



CFG_LED



RECFG_KEY



A Retro Ready Console Powered by FPGA
Shenzhen Sipeed Technology Co.,Ltd.

Sheet: /FPAG_CONFIG/
File: FPGA_CFG.kicad_sch

Title: Tang_138K_Console_V32001C

Size: A4 Date: 2025-04-24

KiCad E.D.A. 9.0.1

Rev: 1.3

Id: 19/17